

CS 2300 HW4

1) Prove that: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Let us say that there exists an element x such that:

$$x \in A \cap (B \cup C)$$

It means that $x \in A$ and $x \in (B \cup C)$ or to be precise $x \in B$ or $x \in C$.

If $x \in B$ and $x \in A$, then $x \in (A \cap B)$ and if $x \in C$ and $x \in A$, then $x \in (A \cap C)$.

Thus we can say that either $x \in (A \cap B) \cup (A \cap C)$.

Since x is an arbitrary element we can say that:

$$A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$$

Let us say that there exists another element y such that:

$$y \in (A \cap B) \cup (A \cap C)$$

It means that $y \in (A \cap B)$ or $y \in (A \cap C)$ or to be precise $y \in (A \cap B)$ or $y \in (A \cap C)$.

If $y \in (A \cap B)$ then we can say that $y \in A$ and $y \in B$, which implies that $y \in B \cup C$ since y already belongs to B . Also, since x is an arbitrary element, we can say that $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$.

Since, we have shown for both x and y that both $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ and $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$, through which we can say that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

2) Prove: $A - (B \cup C) \neq (A - B) \cup (A - C)$

Let us first examine the expression that we need to prove. Let us look at the L.H.S that is:

$$A - (B \cup C)$$

So over here, we need an element that belongs to A but not to B and C. Now, let's examine the R.H.S:

$$(A - B) \cup (A - C)$$

So over here, for the first part that is $(A - B)$, we need an element that belongs to A and not B and for the second part that is $(A - C)$, we need an element that belongs to A and not C.

Now let's move on to the counter example.

Let us assume that:

$$A = \{x, y, z\}$$

$$B = \{y\}$$

$$C = \{z\}$$

Now if we compute the L.H.S, we can say that:

$$A - (B \cup C) = \{x\}$$

Now if we look at R.H.S, we can say that:

$$(A - B) = \{x, z\}$$

$$(A - C) = \{x, y\}$$

$$(A - B) \cup (A - C) = \{x, y, z\}$$

Thus, we can see that $L.H.S \neq R.H.S$

Hence, we have proved by providing a counterexample that $A - (B \cup C) \neq (A - B) \cup (A - C)$.

3) Prove: $(A - C) \cap (C - B) = \Phi$.

Contradiction rule

Let us assume that there exists an element such that:

$$x \in (A - C) \cap (C - B)$$

So that means, $x \in (A - C)$, so $x \in A$ and x does not belong to C .

Similarly, we can say that x belongs to C but does not belong to B .

But here we have a contradiction, since according to the first sentence, x does not belong to C and according to the second one it belongs to C and it cannot happen at the same time.

Hence since our assumption $x \in (A - C) \cap (C - B)$ is a false statement, we can say there exists no such element such that it is common between $(A - C)$ and $(C - B)$.

Therefore, $(A - C) \cap (C - B) = \Phi$.

4a) Prove: If $(A \cup C) \subseteq (B \cup C)$, then $A \subseteq B$

Example:

Let us assume three sets A , B and C such that:

$$A = \{x\}$$

$$B = \Phi$$

$$C = \{x, y\}$$

Now, $(A \cup C) = \{x, y\}$ and $(B \cup C) = \{x, y\}$, so since $\{x, y\}$ is a subset of itself then we can say that the condition $(A \cup C) \subseteq (B \cup C)$ holds true.

Since A has one element (x) and B does not have any element, we can clearly say that A is not a subset of B .

Thus, we can say that if $(A \cup C) \subseteq (B \cup C)$, then $A \not\subseteq B$ which disproves the original statement.

b) Prove: $(A \cap C) \subseteq (B \cap C), A \not\subseteq B$

Example:

Let us assume 3 sets A, B and C such that:

$$A = \{x, y\}$$

$$B = \{y\}$$

$$C = \{y, z\}$$

Now, $(A \cap C) = \{x, y\} \cap \{y, z\} = \{y\}$ and for $(B \cap C) = \{y\} \cap \{y, z\} = \{y\}$.

Since $\{y\}$ is a subset of $\{y\}$ the condition $(A \cap C) \subseteq (B \cap C)$ becomes true.

Now since A contains x and y and B only contains y we can clearly say that A is not a subset of B.

Thus, we can say that if $(A \cap C) \subseteq (B \cap C)$, then $A \not\subseteq B$, which disproves the original statement.

5) Prove: If $(A \cup C) \subseteq (B \cup C)$ and $(A \cap C) \subseteq (B \cap C)$, then $A \subseteq B$.

Let us assume an element x such that:

$$x \in A$$

and we need to show that x also belongs to B.

So, in order to that let us assume 2 cases where in the first case $x \in C$ and in the second one it does not belong to C

Case 1: $x \in C$

Since $x \in A$ and $x \in C$ we can say that $x \in (A \cap C)$, and since we have been given that $(A \cap C) \subseteq (B \cap C)$, so that means $x \in B$ and $x \in C$.

Case 2: x does belong to C .

Since $x \in A$ and it does not belong to C , we can say that $x \in (A \cup C)$ and similarly $x \in B$ in $(B \cup C)$ since x does not belong to C .

Since, in both the cases $x \in A$ and $x \in B$, we can say that $A \subseteq B$.

6) Prove: $A \Delta B = (A \cup B) - (A \cap B)$

Let x be an element such that:

$$x \in A \Delta B$$

Since, we already know that:

$$A \Delta B = (A - B) \cup (B - A)$$

So, we can say that $x \in (A - B) \cup (B - A)$, then in that case $x \in A$ and does not belong to B for $(A - B)$ and $x \in B$ and does not belong to A in $(B - A)$.

So, from this case we can say that $x \in A$ or $x \in B$, so we can say that $x \in (A \cup B)$ and x does not belong to $(A \cap B)$, since x belongs to only A or B .

Therefore, we can say that $x \in (A \cup B) - (A \cap B)$

Thus, $A \Delta B \subseteq (A \cup B) - (A \cap B)$.

Now let us assume that if $x \in (A \cup B) - (A \cap B)$, then $x \in A \Delta B$.

This means that in x belongs to $(A \cup B)$, so x also belongs to either A or B , and does not belong to $(A \cap B)$, we can say that x does not belong to A and B at the same time.

This means that x only belongs to A or it only belongs to B , so we can say that $x \in (A - B)$ or $x \in (B - A)$. So, $x \in (A - B) \cup (B - A)$, which proves that $(A - B) \cup (B - A) \subseteq A \Delta B$.

Thus, by providing these 2 cases we can say that $A \Delta B = (A - B) \cup (B - A)$, as both the L.H.S and R.H.S are subsets of each other.

7) Prove: $f(n) = 3n + 8$ is one-to-one

A function is said to be one-to-one when for all positive integers a and b , we have:

$$f(a) = f(b) \Rightarrow a = b$$

Let us assume that for some positive integers a and b , we have:

$$\begin{aligned} f(a) &= f(b) \\ 3a + 8 &= 3b + 8 \\ 3a &= 3b \\ a &= b \end{aligned}$$

Since we have shown that $f(a) = f(b)$ only when $a = b$, so we can say that $f(n)$ is one-to-one.

A function is said to be onto when a function for every positive integer y , when there exists a positive integer n such that:

$$f(n) = y$$

This means that for every y we need to check for n ,

$$\begin{aligned} y &= 3n + 8 \\ n &= (y-8)/3 \end{aligned}$$

For $f(n)$ to be onto, n must be a positive integer and it must satisfy the above equation.

Counterexample

Consider $y = 10$. If the function was onto, there should be some positive integers n such that:

$$\begin{aligned} 10 &= 3n + 8 \\ 2 &= 3n \\ \frac{2}{3} &= n \end{aligned}$$

Since, n didn't turn out to be an integer we can say that $f(n)$ is not onto.

Thus, by looking at the above examples we can say that $f(n)$ is one-to-one but not onto.

8) Prove: f is onto and it is not one-to-one

A function is considered to be onto when for every element in the co-domain $P(n)$ has at least one pair (A,B) such that:

$$f(A,B) = A \cap B = C, \text{ for every } C \in P(n)$$

In order to show that f is onto let us assume that for any subset $C \subseteq N$, there exists a pair (A,B) such that:

$$A \cap B = C$$

Let us assume that:

$$A = (C \cup D), \text{ for some } D \subseteq N$$

$$B = (C \cup E), \text{ for some } E \subseteq N$$

From here we can say that $C \subseteq A$ and $C \subseteq B$, so we can say that:

$$\begin{aligned} A \cap B &= (C \cup D) \cap (C \cup E) \\ A \cap B &= C \end{aligned}$$

Thus, we can say that f is onto since $f(A,B) = C$, for any subset $C \subseteq N$.

A function is onto if:

$$f(A,B) = f(X,Y) \Rightarrow A = X \text{ and } B = Y$$

This means that $(A \cap B) = (X \cap Y)$, then $A = X$ and $B = Y$.

Let us assume the following subsets of natural numbers N :

$$A = \{5,6,7\}$$

$$B = \{6,7,8\}$$

$$X = \{6,7,9\}$$

$$Y = \{6,7,10\}$$

Thus:

$$A \cap B = \{6, 7\}$$

$$X \cap Y = \{6, 7\}$$

Therefore, we can say that the function is not one-to-one as $A \cap B = X \cap Y$, but $(A, B) \neq (X, Y)$.

So, from the above 2 cases we can say that the function is not one-to-one, but it is onto.

Hence proved.